

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**Regular/Supplementary Summer Examination – 2025****Course: B.Tech.****Branch: Civil Engineering****Semester: VII****Subject Code & Name: BTCVE705B_Y20 / Engineering Economics****Max Marks: 60****Date: 25-07-2025****Duration: 3 Hr.****Instructions to the Students:**

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	Which of the following best explains the "time value of money"?	L1/CO1	1
	A) Money loses value over time due to inflation. B) Future money is more valuable than present money. C) A sum of money today is worth more than the same sum in the future. D) Interest rates do not affect money's value.		
2	Which of the following is considered an intangible factor in engineering economics?	L1/CO1	1
	A) Raw material cost B) Labor wages C) Brand reputation D) Equipment maintenance cost		
3	Which of the following is NOT a type of cost?	L1/CO1	1
	A) Fixed cost B) Variable cost C) Depreciation cost D) Elastic cost		
4	What does a cash flow diagram represent?	L1/CO2	1

	A) Temperature changes over time	B) Income and expenses over time	C) Interest rates only	D) Project timelines		
5	In Present Worth Analysis, all cash flows are converted to their value at:				L1/CO2	1
	A) Future date	B) Midpoint of project	C) End of project	D) Present date		
6	Which method is used to determine the rate at which net present value becomes zero?				L2/CO2	1
	A) Annual Worth Method	B) Future Worth Method	C) Rate of Return Method	D) Break-Even Analysis		
7	Which of the following is NOT a method of equipment acquisition?				L1/CO2	1
	A) Purchase	B) Lease	C) Borrow	D) Rent		
8	Which estimate uses historical data and statistical models for cost prediction?				L2/ CO3	1
	A) Detailed estimate	B) Parametric estimate	C) Conceptual estimate	D) Preliminary estimate		
9	Depreciation is:				L1/ CO3	1
	A. A method of calculating taxes	B. The increase in value of an asset over time	C. The allocation of the cost of a tangible asset over its useful life	D. Not applicable in engineering economics		
10	Which of the following is a source of long-term finance?				L1/ CO3	1
	A. Petty cash	B. Debentures	C. Trade credit	D. Credit card		
11	Which financial statement shows the company's financial position at a specific point in time?				L1/ CO3	1

	A. Income Statement	B. Cash Flow Statement	C. Balance Sheet	D. Budget Sheet		
12	Working capital is defined as:				L2/ CO3	1
	A. Long-term loans + Fixed assets	B. Current assets - Current liabilities	C. Total assets - Total liabilities	D. Profit - Taxes		
Q. 2	Solve the following.					12
A)	If the interest rate is 6% per annum, how much would you have to invest today to receive Rs 1500 after 10 years?				L3 /CO2	6
B)	Discuss significance of engineering economics.				L2 /CO1	6
Q.3	Solve the following.					12
A)	Explain Uniform series of payments with example.				L2 /CO2	6
B)	Given the cash flow diagram of a project, how would you determine the benefit/cost ratio of the project? What would be the significance of the benefit/cost ratio in determining the economic viability of the project?				L4 /CO2	6
Q. 4	Solve Any Two of the following.					12
A)	Explain the concept of Replacement Analysis and its importance in project decision-making. Provide a scenario where Replacement Analysis would be beneficial.				L3 /CO2	6
B)	Discuss buy, rent and lease options with example				L3 /CO2	6
C)	What are the financial accounting principles? Write a short note on financial statements.				L2 /CO3	6
Q.5	Solve Any Two of the following.					12
A)	Analyse the impact of different depreciation methods on investment evaluation in the presence of inflation.				L4 /CO3	6

B)	Describe budgeting and budgetary control.	L2 /CO3	6
C)	Explain the diverse sources of finance available to businesses. Compare and contrast equity financing, debt financing, and securities as means of raising capital.	L4 /CO3	6
Q. 6	Solve Any Two of the following.		12
A)	Explain the factors governing selection of appropriate PPP model.	L3 /CO2	6
B)	Explain International Financing and FIDIC.	L2 /CO3	6
C)	Define a Balance Sheet and explain its importance in financial reporting. Highlight the fundamental components typically found on a Balance Sheet.	L2 /CO3	6
	*** End ***		